26900141 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Core Course (CC) - MG5CCRBCA300 - COMPUTER NETWORKS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Define a digital signal. [K] / [CO1, CO2]
2. List any four networking devices. [K] / [CO1]
3. Differntiate Circuit switching and Packet Switching [U] / [CO1]
4. Explain the significance of NAK control message. [U] / [CO2]
5. Explain CSMA. [U] / [CO2]
6. Mention any two functions of the Data Link Layer. [U] / [CO2]
7. A computer knows the IP address of the destination but not its physical address. Which protocol do address resolution? [A] / [CO2]
8. Define network latency. [K] / [CO3]
9. Explain two components of an HTTP request message. [U] / [CO3]
10. Explain network management. [U] / [CO3]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **3** questions

Each question carries **5** marks

11.	Describe with examples how computer networks are used in real-time applications like video conferencing and online gaming.	[U] / [CO1]
12.	Explain mesh topology with a diagram, and discuss its advantages and disadvantages.	[U] / [CO1]
13.	Explain the IEEE standards used for Wired LANs and describe their major characteristics.	[U] / [CO2]
14.	Describe the IPv4 addressing scheme.	[U] / [CO2]
15.	Give the primary objectives of network monitoring.	[K] / [CO3]

[5x3 = 15]

Part C

Essay Questions

Answer any **1** questions

Each question carries **15** marks

16.	With the help of neat diagram explain the functions of each layer in OSI reference model.	[K] / [CO2]
17.	Explain the major TCP congestion control techniques in detail. Discuss the working of Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery process.	[U] / [CO3]

[15x1 = 15]